LOT HW-001 ??? 00-PLAN-AND-ROADMAP

LOT Computer (HW-001) — Plan, Analysis, Roadmap

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TL;DR (compressed)

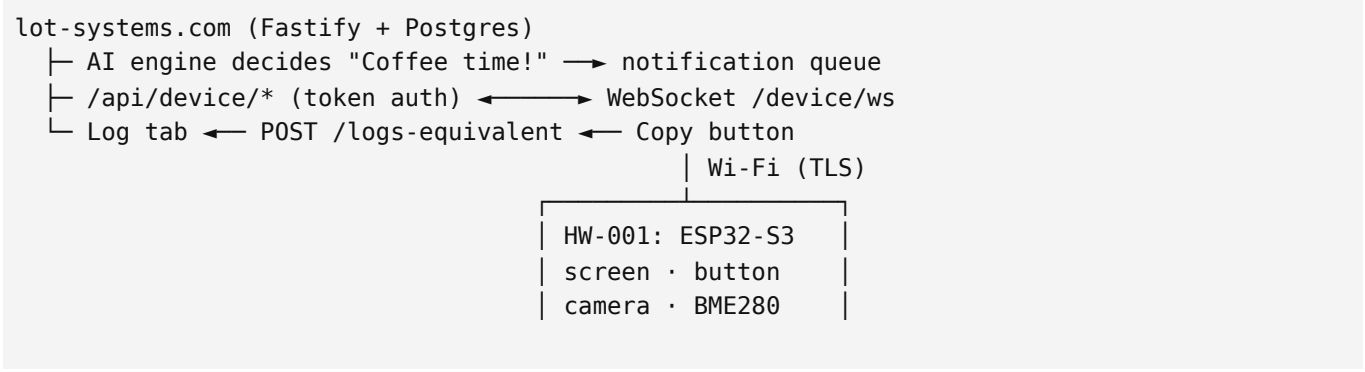
- **Product:** a 40 × 40 mm stainless-steel “pager” that receives autonomous AI notifications from lot-systems.com (“Coffee time!”), shows them on a small screen, and sends a **Copy** button press back to the site’s **Log** tab.
- **Buildable now:** ESP32-S3 + OV2640 camera + 1.54” IPS + BME280 weather sensor + Qi charging. All off-the-shelf.
- **Biggest risk #1 — 5 mm thickness.** Camera + screen + battery + Qi coil + PCB does not fit in 5 mm with standard modules. Realistic v1 is 9–10 mm; 5 mm needs a custom flex/bare-die design (v2).
- **Biggest risk #2 — metal blocks radio and Qi.** A sealed stainless body kills Wi-Fi/BLE and wireless charging. Both need a non-metal window (glass, ceramic, or PEEK). This changes the “one side polished stainless” idea: the polished face stays solid, and the charging and antenna window goes on the other side or in a slit.
- **Backend gap:** `POST /logs` exists but needs a browser session (`req.user`). Devices need token auth and a push channel. Spec in `02-API-CONTRACT.md`.
- **Cost (estimate, ±30%, replace with quotes):** about $95–125 per unit at 100 units, so about $10–13k for the run plus about $1–2k one-off costs.
- **Timeline:** about 22–26 weeks from today to 100 units shipped. Gate: the thickness decision (Phase 1).

1. Your 19 requirements mapped

#	Requirement	Plan	Doc
1	PCBWay	PCB fab + turnkey assembly; stainless parts via PCBWay CNC service (quote-based)	05
2	Pager-like AI notification	Server pushes `notify` events over WebSocket/MQTT; device beeps/vibrates and shows text	02, 03
3	2-part stainless body	316L top and bottom shells, laser-welded or 4× M1.2 screws, with gasket	04
4	40×40×5 mm silver square	Target dimension; v1 = 40×40×9.5 mm, v2 = 5 mm (see	04

#	Requirement	Plan	Doc
		04)	
5	Camera	OV2640 (2 MP) v1; OV5640 optional	01
6	LOT API connector	Device token + `/api/device/*` endpoints	02
7	PDF manuals	Markdown → PDF build (`docs/hardware/build-pdf.sh`)	all
8	Compress info per session	Every session ends with a TL;DR + decisions + open items (Section 8)	00
9	Firmware documents	`03-FIRMWARE-SPEC.md`	03
10	Software to connect with firmware	Web flasher (ESP Web Tools) + `/device` page on the site	02, 03
11	Separate documents	00–05 split by audience	index
12, 19	Charger / wireless charger	Qi receiver (5 V, ~1 W) + Li-Po charger IC; Qi works only through a non-metal window	01, 04
13	100-unit run	Pilot of 10, then EVT/DVT, then the 100 run	05
14, 15	Weather sensor, AI-grade off-the-shelf sensors	BME280 (T/RH/P) in v1; BME688 (gas/air quality, runs Bosch BSEC AI) as an upgrade; IMU optional	01
16	Button “Copy” → Log tab	Press sends `POST /api/device/event {type:"copy"}`; the server writes `Log{event:"device_copy"}`	02
17, 18	One side polished steel; other side camera + screen + button	Front = camera, screen, button. Back = mirror-polished solid steel, with a charging window (see risk #2)	04
—	Simple screen for autonomous notifications	1.54” 240×240 IPS, text-only UI	03

2. Architecture



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                                  │ Li-Po · Qi RX      │
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3. Analysis (what the brief does not say yet)

1. **Thickness (5 mm).** Rough stack with standard parts: steel walls 2 × 0.6 mm, screen module 2.5 mm, PCB 0.8 mm, ESP32-S3 module 3.2 mm, OV2640 module about 4–5 mm, battery 3.2 mm, Qi coil 0.5 mm. That is about 14 mm if stacked naively. Side-by-side layout plus a custom PCB gets to about 9–10 mm. 5 mm requires bare ESP32-S3 on a flex PCB, a 2–3 mm camera module, and a 2 mm battery. That is a v2 effort.
2. **RF through metal.** Put the antenna under a non-metal inlay. A 40 mm body makes the antenna small; test early with a mock-up.
3. **Qi through metal.** Eddy currents block charging through 316L. Use a glass, ceramic or PEEK window over the coil, or pogo-pin charging as a fallback.
4. **Camera purpose is undefined.** Options: photo-to-Log, QR scanning, or pairing. Recommend QR pairing and photo-to-Log in v1 (shutter = long-press).
5. **Battery safety.** A Li-Po cell in a sealed steel case needs a protection circuit and UN38.3 testing before shipping; plan for it in the 100-unit run.
6. **Compliance.** Wi-Fi, Bluetooth and Qi mean FCC Part 15 testing (about $5–10k for a full lab). A pre-certified ESP32-S3 module reduces the scope. Needed before any sale; not needed for internal units.
7. **Security.** Per-device token, rotated by the server, TLS pinning, no secrets in firmware images.
8. **“Autonomous notifications”.** The site needs a rules or AI trigger (scheduler → AI → notification), not just a manual send.

4. Roadmap

Phase	Weeks	Output	Gate
0. Plan (this doc)	now	Docs 00–05	Approve BOM and risks
1. Prototype on dev kits	1–4	Breadboard on ESP32-S3 dev kit plus screen, buttons, BME280, camera; firmware talks to a staging API	Notification shown; Copy appears in Log
2. Backend	2–6 (parallel)	Device auth, notification queue, WebSocket, Log event, `/device` page	E2E on the dev kit
3. Industrial design	3–8	CAD for 2-part body, thickness decision (9.5 mm v1 vs 5 mm v2), RF/Qi window test	Signed-off drawings
4. EVT PCB	6–12	KiCad design, 5–10 boards via PCBWay, first stainless samples	Boards boot, charge, connect
5. DVT	12–18	10–20 units, drop/seal/thermal tests, firmware OTA, PDF manuals	Pass criteria in 05

Phase	Weeks	Output	Gate
6. PVT / 100 run	18–26	100 units, QC sheet, UN38.3 shipping docs	100 units shipped

5. Decisions needed from Vadik

1. v1 at 9–10 mm, or hold for 5 mm? (Recommend: v1 at about 9.5 mm.)
2. Charging window material: glass, ceramic, or PEEK? (Recommend PEEK for v1.)
3. Notification transport: WebSocket (recommended) or MQTT?
4. Camera role: QR pairing and photo-to-Log?
5. Will units be sold, or internal only? That decides FCC and UN38.3 scope.
6. Budget approval of about $13k for the 100 run.

6. Limits of this session

- lot-systems.com, brand.lot-systems.com and institute.lot-systems.com were **unreachable** from this environment (DNS failure), so brand rules, the About page and the CQGS page were **not read**. Brand tone in these docs follows only what is in the repo. Please re-run with network access or paste those pages.
- All prices and thickness figures are engineering estimates, not quotes. Component links are manufacturer or vendor pages; stock and price were not verified.

7. Sources

- PCBWay quote page (assembly is quote-based, BOM needs MPNs)
- Adafruit MEMENTO camera board (ESP32-S3 + 1.54” 240×240 + camera reference)
- DFRobot FireBeetle 2 ESP32-S3 + OV2640 on DigiKey (9-week lead time seen)
- Waveshare ESP32-S3-LCD-2 (ST7789 + camera interface)

8. Session log (append one block per session)

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2026-10-19 S1  Plan v0.1 written. Docs 00–05 created. Sites unreachable.
               Decisions pending: Section 5. Next: approve BOM, start Phase 1 + 2.
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